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Explainable COVID-19 Detection from Chest X-Ray Images Using Deep Learning and Tranfer Learning

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Abstract*—Abstract—Chest radiography is a common method of evaluating respiratory system abnormalities. may be difficult to interpret due to respiratory disease, Prior to deployment, image-based AI tools need to be thoroughly tested. fore any medical use. This paper introduces a suggested explainable A deep-learning system to categorize chest X-ray images as COVID-19, Normal and Pneumonia categories. The workflow uses transfer learning with DenseNet121 and EfficientNetB0, Uses uniform image pre-processing and data augmentation, and Gradient-weighted Class: Activation Mapping (Grad-CAM). Evaluation is planned using: For this purpose accuracy, class-wise precision and recall, macro-F1, confusion are used. matrices, sensitivity, specificity and area under (one-versus-rest) When appropriate, the receiver operating characteristic curve. The This dataset has been derived from the publicly-described V7 Labs COVID- A set of X-ray images and their associated classes, count of usable images and pa-. Identifiers, duplicates, and license verification must be done from tient. Before training the downloaded files are erased. No performance result The total of the numbers written in this paper is claimed. Initially measured and submitted. Many of them are the principal contribution, An educational pipeline that is reproducible and can incorporate the classification of, and visual explanations, whilst explicitly taking on board In addition to the difficulties of public medical- image datasets. Keywords—CNN, Chest X-rays, COVID-19 A wide variety of efficient networks have been developed. network, transfer learning, DenseNet121, EfficientNetB0, Grad- Explainable Artificial Intelligence (XAI).

***Index Terms*—component, formatting, style, styling, insert.**

I. Introduction

Clinically, respiratory diseases can have similar signs or symptoms, and chest X-ray (CXR) images are one of the imaging modal- Lung abnormalities used to be evaluated by the use of the above-mentioned ities. Deep learning methods can identify visual characteristics on x-rays, and aid in imaging support classification research. But with a high classification score However, it is not necessarily reliable on a small sample size. Performance when used in a new hospital, population or imaging environment. The COVID-19 image classification studies have been emphasized. This is because of method-ological weaknesses, such as data leakage, confound- The two shortcut learning are ing, [8], [9]. This is a project on an educational system that classifies CXR images into three categories: COVID-19, normal, and pneumonia. It makes a comparison between DenseNet121 and EfficientNetB0. using transfer learning and uses Grad-CAM to visualize image Areas that affect a forecast. DenseNet connects feature The gated dense skip connections are used to support the feature. network width up to 90-fold and its in- network thickness to 11Coordinate the resolution [5]. Grad-CAM uses sort of coarse filtering in the convolutional layers, and then gradients are pushed out of the convolutional layers to form a coarse A mapping of a class on a local level [6]. The goals are to: (1) inspect and record the chosen Generate a dataset and its labels; (2) implement a reproducible preprocess- ing and splitting pipeline; (3) fine-tune two pretrained CNN mance of the system based on these architectures; and (5) compare the perfor- (1) write and record; (2) record predictions; (3) show selected predictions, and (4) visualize selected predictions using Grad-CAM. This is a research prototype and not a clinical diagnostic. device

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II. Related Work

Com-ponential networks, known as convolutional neural networks (CNNs), have caught the public's attention recently. Hiring someone to do medical image analysis would be very expensive, as they would need to be trained in a hierarchical manner. Convert images to visual representations from image pixels. Litjens et al. survey Explain some deep-learning applications in medical imaging and Suitable validation [7] is important. Kermany et al. Showed how to use deep learning based on images for medical diagnosis Chest radio- and optical coherence tomography (OCT) techniques were used to analyze tasks. On graphs such as pneumonia classification [2] . . . . . Rajpurkar et al. introduced CheXNet, a deep-learning system for pneumonia chest radiographs [3] have been used for detection. The COVID-19 Image Data Collection gathered radio- Use of graphic images from open sources to aid research during the pandemic [1]. The V7 Labs data set repository is a collection of federated data sets. The images in this section are from a pneumonia and provide examples for COVID-19. consideration of dataset and documents annotations erations [11]. These are mixed-source collections that are useful for:

While capable of prototyping, can introduce source-specific differences un- related to disease. Roberts et al. suggest that careful design is required. and reporting models for imaging the COVID-19 [8]. DeGrave et al. demonstrated the possibility of a model using shortcuts instead of med- In the field of radiograph classification in COVID-19, important signals are being detected at the right place and time. [9]. DenseNet121 is based on densely connected convolutional EfficientNet offers a compound-scaling model, whereas networks [4] offer a different approach. CNN approach for CNN design [5]. Grad-CAM is widely used to However, for every individual class, they can create a visual explanation [6]. The heatmap does not prove that a model depends on and is using Based on clinically sound data. The proposed study is a combination of these techniques in one pipeline and highlights the transparency of the keywords. Handling of data sets and careful interpretation.

III. Theoretical Background

A. Convolutional Neural Networks

A CNN takes an image as a tensor as its input. A convolutional layer Uses learnable filters over the local regions of an image to generate feature maps. For an input image Image or feature map X and filter K, a simplified two-dimensional convolution is given by

$$Y(i,j) = \sum_m \sum_n X(i-m, j-n)K(m,n) + b, \quad (1)$$

where b is a bias term. Early layers are likely to learn edges and Deeper layers can then represent more intricate patterns, and textures can be represented in the shallower layers. A rectified linear unit (ReLU) is given by

$$\mathrm{ReLU}(x) = \max(0, x). \quad (2)$$

Pooling or strided convolutions will decrease the resolution of the space and computational cost. Not strictly speaking a classification head, but one that is able to aggregate learned. Displays and calculates class scores.

B. Transfer Learning

The transfer learning method is based on the premise that a model trained on one data set can be adapted to another by transferring its learned weights from the first to the second. Has a large source domain and adapts it to a target task. In In feature-extraction mode the pretrained backbone is frozen and Only new classification head is trained. During fine-tuning, some or all of the backbone layers are unfrozen and updated with a smaller learning rate. The images can be pre-trained on ImageNet. ability to be used for useful generic visual features, but their appropriateness for Radiographs must be determined by trial and error. Input-channel For example, the handling, resizing images, and the preprocessing they are expected to receive. All of the architectures should be in accordance with the selected implementation.

C. DenseNet121 and EfficientNetB0

DenseNet links all the layers together with layers that follow in a dense block. The feature maps from the previous layers are Later layers will be concatenated to this layer's input, encouraging feature reuse. and enhancing the gradient flow [8]. Dense connectivity can require more memory due to intermediate feature Maps are to be kept.

EfficientNet uses compound scaling to coordinate network Measure the depth, width and resolution of images, not scaling one. dimension alone [9]. EfficientNetB0 is a small A baseline that can be viable for a limited GPU resource. Model performance is dependent on the quality of the data, training protocol, Its names are not enough to ensure the architecture; and evaluation design; Determine which model will be more successful.

On this topic we consider the output probabilities and loss. The model yields logits for a three-class task. z_1, z_2, z_3. They are then passed through the softmax function to be converted to normalized scores:

$$p_k = \frac{\exp(z_k)}{\sum_{j=1}^{3} \exp(z_j)}, \quad k \in \{1,2,3\}. \quad (3)$$

These scores are not necessarily, but are instead, model probabilities: calibrated probabilities of disease were . Let y_k be a one-hot label, The categorical cross-entropy is given by:

$$\mathcal{L} = -\sum_{k=1}^{3} y_k \log(p_k). \quad (4)$$

To train model parameters, use the gradients of the loss, computed typically using backpropagation and optimized by stochastic gradient-based methods.

D. Dataset and Data Preparation

The dataset and data preparation process are described below.

The project is based on the dataset repository for COVID-19 X-ray images provided by V7 Labs.The project is built on top of the COVID-19 X-ray dataset repository provided by V7 Labs. as the starting point [**?**]. Its documentation describes Images taken from a pneumonia and COVID-19 images were used for comparison. dataset, notes that the repository has lung segmentation masks. There are 517 COVID-19 reported in the repository description. Data from 5,863 images of pneumonia and 2,304 images from a Mendeley pneumonia source; These numbers should not, however be considered final three-class training counts. The real disk files downloaded, Duplicates and exclusions, labels, and metadata need to be Be checked and documented prior to training. The normal class must be identified from verified metadata and not be inferred. Without proof or evidence from filenames.

A dataset audit should be used to document the number of images that are usable. By class, source, file format and exclusion reason. Duplicate There should be a duplicate and/or patient identifiers. be checked. If identifiers are available for the patient, the dataset should be separated by patient such that

images from the same patient go in the same folder. cannot be used in the training and test sets. If patient-level If there are no identifiers, then this limitation should be explicitly described. explicitly. The fixed random seed should be noted for purposes of supporting reproducibility.

Images are transformed to desired color space and resized to the desired size. the input resolution that each model expects, and normalized With the selected pre-trained model's preprocessing function. Data augmentation can involve slight rotations, translations, and or mild contrast changes (only on the training set) Augmentations cannot be used to make anatomy unrealistic or change the clinical label. Validation and Test images should NOT be given. random augmentation. Class weights or balanced sampling may be taken into account if the class distribution is significantly different from what is expected. Imbalanced; the chosen strategy should be recorded.

IV. Proposed Methodology

The proposed pipeline includes dataset auditing, and We have data preprocessing, data splitting, model training, evaluation etc. and explanation.

A manifest is first created with the following content: path of the image, label, Source and patient identifier (if available). The manifest All files are checked for invalid, duplicate, and label information. inconsistencies. The data are then split into training, Before selecting a model, validate them and test them through partitions. The test set is only used for the last comparison.

Both models are pre-trained and they both have a new 3 class classification head. The following two-stage procedure is recommended: Freezes backbone, and trains the classification head, followed optionally by unfreezing selected upper layers for. fine-tuning. The learning rate, batch size, optimizer, number The time of epochs, early-stopping rule and checkpoint selection. validation set should be used to choose the criterion. recorded. The same data split and similar training methods were used. If possible, budgets should be used for both models. Prespecified criteria are used to choose the best checkpoint. validation loss or macro-F1.

The final checkpoints are assessed after training on the test set that is not used. Predictions, class probabilities, and frequencies: errors are stored to enable the generation of metrics and Grad-CAM images. be reproduced. The system can be exposed via a Gradio model that takes an image as an input and gives the predicted class as an output. and model scores, and shows a Grad-CAM overlay. Images to be uploaded should not be stored or transmitted beyond. without permission, enter the intended environment. appropriate safeguards.

V. Explainability with Grad-CAM

Class Activation Mapping (Grad-CAM) Generates a coarse heat map showing areas that affect A chosen class grade for the selected class. Let A^k be the feature map k. The score of the convolutional layer is denoted by z^c and y^c denotes the score for class c. is defined as the value of the feature map k that is used to weight the importance of the network. obtained by globally averaging the gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \tag{5}$$

where the Z are the number of spatial positions. The The class-specific Grad-CAM map is given by

$$L_{\text{Grad}-\text{CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right). \tag{6}$$

The heatmap will be resized and placed on top of the original. radiograph. ReLU is positive on the selected: class in this formulation. The examples below are for each model. The furniture in all three classes should be checked, especially the correct predictions and errors. A map showing the color gradient of borders, may be found on labels, image corners, or medical devices. shortcut or confounder. Visual inspection is qualitative - the amount of waste is not measured. Should NOT be used to imply clinical validation.

VI. Evaluation Metrics

In a large set of predictions, accuracy is the percentage of accurate predictions. are correct. If it is a binary classification problem, then it is: defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{7}$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives. true negatives, false positives and false negatives, respectively.

For the rest of the class, this is considered a positive class. A class is defined as precision and a recall is defined as classes.

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}. \tag{8}$$

F1 Score is the harmonic mean of precision and recall:

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{9}$$

Sensitivity = recall and specificity is =

$$\text{Specificity} = \frac{TN}{TN + FP}. \tag{10}$$

In the case of three class task, metrics must be class-wise. A one-versus-rest interpretation was used by reported. Macro-F1 Makes equal weightage to the F1 score of each class. A confusion matrix should also be provided and, if reliable class scores One versus rest receiver operating are available. It should be the characteristic area under the curve (ROC-AUC). be reported. The metric case is not defined for a class with Must be handled and disclosed, no positive predictions. Definitions and implementations of the metric can be supported. The documentation for using scikit-learn and the relevant literature. [20].

VII. Experimental Setup

The environment for the planned exercises is Google Colab and Python. If available, use a GPU. TensorFlow/Keras or PyTorch can be used. Must be used, the framework and the versions of it should be selected. be recorded for the last experiment. The runtime GPU The following variables were selected for the analysis: model, library versions, input resolution, and batch size. Best accuracy with optimizer, learning rate, epochs, and early stopping. The configuration, class weighting strategy and random seed. should be documented.

For DenseNet121: the same data partition for training/testing should be followed. and EfficientNetB0. The trained weights and prediction outputs are displayed. and training logs should be stored for reproduction.

VIII. Results and Discussion

This report makes no claims as to experimental results. The comparison of the models should be made after The selected checkpoints in the held-out test set. Performance values should be determined from the Experiments and not estimated from training accuracy. Or plagiarized from other sources and used in such a way as to appear as if they were original. by this project.

IX. Limitations, Ethics And Bias

Public datasets can be made up of images from various institutions, devices, acquisition protocols and patient populations. A model may utilize text marks, borders, portable-device Artifacts, patterns or not disease-related. lung findings. These shortcuts can cause the performance to be exaggerated and reduce generalization [15], [16]. The disproportion.The imbalance of classes. The absence of external validation can impact test sets, as well, reported metrics.

The study should include information about the source of the datasets, any legal restrictions on their use and the copyright notice, The exclusion rules and, where known, restrictions at the patient level. Public availability does not allow for free flowing. The terms surrounding each image; redistribution of each image source has to be obeyed. Grad-CAM cannot draw causal Reason or correctness in an argument. The application is designed to be used for For educational research only and must not be used to diagnose, Make treatment decisions, or triage. For clinical use would require Independent External Validation, Expert Review, Governance, privacy protection, assessment of regulations.

X. Conclusion and Future Work

In this paper, a proposed explainable chest X-ray is described. Transfer learning of classification pipeline with NormalXception, COVID-19Xception for COVID-19 and normal. pneumonia classes. To make model Grad-CAM easier to understand and use, it includes the following: behaviour more easily monitored, and a multi-metric assessment of the behaviour. plan relates to performance for each class. The credibility of the final results depends on verified labels, leakage-free splitting, reproducible training and transparent reporting.

Future work should test on external data sets, evaluate calibration solve for and measure explanation stability, obtain expert review of model explanations. Until these steps The system is considered as an educational system when are completed. A prototype, not a clinical tool!

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The authors would like to express their gratitude to the creators and contributors of the following Open Access books and resources. A component of the publically available chest X-ray database employed for this study. They have worked to make medical imaging data available to The use of deep learning to detect COVID-19 was supported by this research. classification and explainability.

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